

Pentana Vehicle Sales

Dealer Land-Size Enrichment — East-Coast Progress Report

Synogize Internal Technical Report

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1 Executive Summary

This report documents the dealer land-size enrichment built on top of the curated Pentana dealer reference. The goal is to link verified dealer addresses to official cadastral parcel areas, adding a physical-site layer to dealer analysis.

The workflow is now proven across the full east coast. As of this update, **1,339 of 1,530 east-coast dealer rows** (87.5%) have parcel-backed land-size values, sourced exclusively from official Australian government spatial data. This reflects multiple recovery passes (B013, B014, manual same-site review) applied after the original pilot, expanding both the row count and coverage rate.

Metric	Value	Notes
East-coast pilot rows (all batches)	1,530	NSW + VIC + QLD MATCHED / POSTCODE_MISMATCH / PARTIAL
East-coast parcel-backed rows	1,339	87.5% coverage
NSW	490 / 535	91.6%
VIC	478 / 542	88.2%
QLD	371 / 453	81.9%
Remaining gap	191	73 geocode review + 68 parcel selection + 50 targeted rerun
QLD	82	Weakest state; priority for next pass
NSW	45	Hard address and frontage cases
VIC	64	Mostly parcel-selection review
Median land size (NSW pilot)	5,327 m ²	
Smallest site	444 m ²	Compact urban showroom

2 Why This Enrichment Matters

The dealer reference workstream established *who* each dealer is. This enrichment establishes *how large* their physical site is.

That creates a new analytical layer that was not previously possible from the DMS data alone:

- **Sales intensity** — units sold per square metre of site area. A dealer selling 800 vehicles per year from 3,500 m² is operating very differently from one selling the same volume from 15,000 m².
- **Capacity and scale benchmarking** — large-format campuses vs. compact urban showrooms have structurally different cost bases, staffing requirements, and throughput ceilings.
- **Network planning** — OEMs and distributor groups can identify which dealers have physical room to expand, add a new brand line, or build a new service facility.
- **Performance context** — apparent under-performance may partly reflect site constraints. A small-footprint dealer in an inner-ring suburb cannot achieve the same throughput as a regional super-site regardless of management quality.
- **Real-estate proxy** — land size $\times$ location gives a rough property value component useful in franchise valuation and M&A analysis.

This enrichment should be interpreted as a benchmarking layer. Parcel area is not the same as usable floor area — a site may include rear vehicle storage, access laneways, or adjoining lots. It is nonetheless the most authoritative and freely available proxy for dealer footprint at this scale.

3 Methodology

3.1 Data sources

Source	Details
NSW Cadastre WFS	<code>maps1.six.nsw.gov.au</code> — official NSW Spatial Services parcel layer. Licenced CC-BY. Point-in-polygon intersection returns <code>shape_Area</code> for the cadastral lot.
NSW Property layer	Used for multi-lot dealer campuses where a single cadastral parcel understates the full site. The <code>Property→PropertyLot</code> join aggregates all constituent lot areas into one site total.
G-NAF	PSMA Australia Geocoded National Address File — the authoritative Australian address-point dataset. Provides building-level coordinates for dealer addresses with no rate limits and no API cost. Used as the primary geocoder.
Same-site inheritance	Where multiple dealer proxy keys share a verified physical address (multi-brand campuses), a confirmed parcel hit is inherited across all keys at the same site under explicit review.

3.2 Exact data flow — how land size is derived

Land size is never read directly from the dealer address. It is derived in up to six steps through official data services. The address is the starting point; the cadastral parcel area is the end point.

Step	Stage	Input / Output
1	Dealer address	Input: curated address string from the dealer reference. Example: <i>519 Princes Highway, Kirrawee, NSW, 2232, Australia</i>
2	Geocoding (G-NAF)	Input: address string. Output: latitude, longitude, match quality flag. G-NAF (PSMA Geocoded National Address File) is used for NSW — not a generic web geocoder. It resolves to building-point level precision.
3	NSW Cadastre WFS	Input: longitude, latitude, geometry type = point. The service performs a spatial intersection and returns: parcel id, lot/plan identifiers, shape_Area . shape_Area is stored as LAND_SIZE_SQM . This is Path A — the automated direct path.
4	NSW Property layer	Used when Path A fails or the site spans multiple lots. Input: official property address text or nearby point. Output: propid for the physical site.
5	PropertyLot join	Input: propid from step 4. Output: all cadastral lot references (cadid) linked to that property.
6	Cadastre lot query	Input: cadid or lot reference from step 5. Output: official area per lot. For multi-lot sites (e.g. Warwick Farm = 11 lots), lot areas are summed. This is Path B — the property-layer path.

In summary, there are two resolution paths and one recovery mechanism:

- **Path A (automated):** address → G-NAF point → NSW Cadastre → area
- **Path B (reviewed):** address → NSW Property → PropertyLot → Cadastre → area
- **Same-site inheritance:** once one site is confirmed via Path A or B, all other dealer proxy keys at the same verified physical address inherit that parcel and area under explicit review

Worked example — Path A: Tynan Suzuki at 519 Princes Highway, Kirrawee NSW 2232. G-NAF returns coordinates at building-point level. The NSW Cadastre WFS intersects that point and returns parcel 2//DP1234567 with **shape_Area** = 6,120. That value is stored as **LAND_SIZE_SQM** = 6120 with **CADASTRAL_CONFIDENCE** = HIGH.

Worked example — Path B (multi-lot): Peter Warren Automotive at 13 Hume Highway, Warwick Farm. The G-NAF point fell on one lot but the full site spans 11 cadastral lots. The NSW Property layer resolved property ID 4396954, which links to 11 lots via PropertyLot. Summing their areas gives 117,528 m². Stored as **LAND_SIZE_SQM** = 117528 with **CADASTRAL_CONFIDENCE** = MEDIUM (property-layer reviewed path).

3.3 Pipeline stages

1. **Candidate preparation** — filter the curated dealer reference to NSW MATCHED and POSTCODE_MISMATCH rows with a valid street address. Assign confidence tier based on match status. Output: 34_dealer_land_size_east_coast_pilot.csv (411 NSW rows).
2. **Geocoding (G-NAF)** — resolve each dealer address to a precise building-level coordinate. Strict review flags classify results as GEOCODED_OK, GEOCODED_REVIEW, or NO_RESULT. Road-level matches are flagged but not auto-accepted.
3. **NSW cadastre lookup** — send each accepted geocode to the NSW Cadastre WFS. A spatial intersect returns the parcel identifier and shape_Area. Result: CADASTRAL_STATUS = PARCEL_FOUND with CADASTRAL_CONFIDENCE = HIGH.
4. **Property-layer site resolution** — for multi-brand campuses where a direct point miss occurs, query the NSW Property layer to identify the full property boundary, aggregate all constituent lot areas, and seed the result as a manual review override.
5. **Same-site inheritance** — for proxy keys at verified shared addresses, inherit the confirmed parcel and area from the seed row. Flagged PARCEL_FOUND_MANUAL_INHERIT at MEDIUM confidence.

3.4 Confidence tiers

Tier	Meaning
HIGH	G-NAF address-point geocode + direct NSW Cadastre parcel hit. Both the coordinate and the parcel are from authoritative official sources.
MEDIUM	Same-site inherited parcel, or property-layer reviewed seed. The parcel data is authoritative, but the address match was reviewed rather than resolved at building-point precision.
LOW	Road-level geocode used for the parcel intersection. The parcel hit is plausible but not confirmed at address-point precision.

4 Recovery Method and Coverage

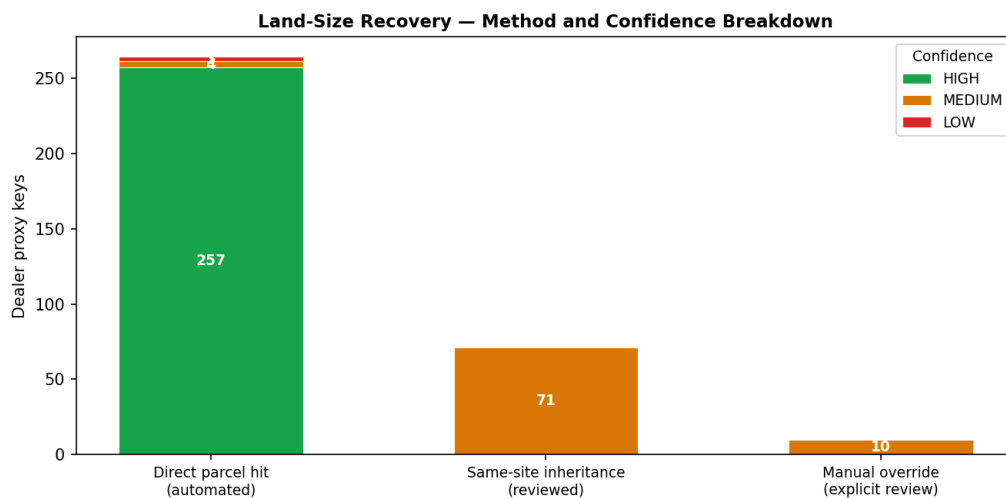


Figure 1: Breakdown of the 345 parcel-backed NSW rows by recovery method and confidence tier. Direct automated hits (left) are predominantly HIGH confidence. Same-site inheritance (centre) is MEDIUM by design — the parcel is authoritative but the address match was reviewed. Manual property-layer seeds (right) are also MEDIUM.

Recovery method	Keys	%	How
Direct parcel hit	264	76.5%	G-NAF point intersected NSW Cadastre WFS
Same-site inheritance	71	20.6%	Reviewed shared-address cluster
Property-layer seed	10	2.9%	Manual NSW Property layer + lot aggregation
Total	345	100%	

The 66 unresolved NSW rows fall into three categories:

Unresolved type	Rows
No geocode result (address not in G-NAF)	48
Review-only geocode (road-level match)	15
Geocoded but no parcel returned	3

These are genuinely hard cases: corner addresses, newly developed sites not yet in G-NAF, or dealer yards on multiple non-contiguous lots. They are not batchable with the current automated pipeline and would require individual manual review.

5 Land-Size Distribution

5.1 Full distribution (log scale)

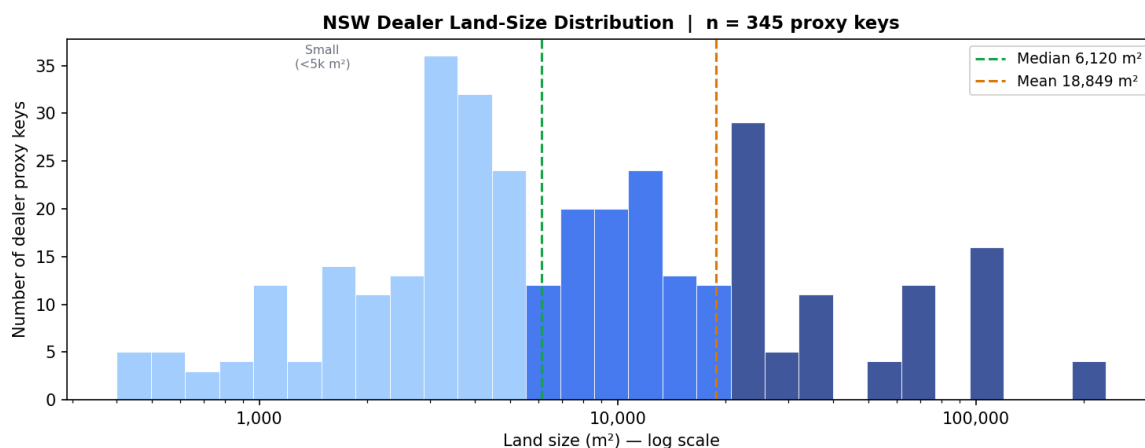


Figure 2: NSW dealer land-size distribution using log-scale bins. A linear x-axis would compress all variation into the first few bins because the data spans nearly three orders of magnitude (444 m² to 209,570 m²). Log bins give equal visual weight to each order of magnitude, making the shape of the distribution readable. Three zones are shaded: *small showrooms* (< 5,000 m²), *mid-size dealerships* (5,000–20,000 m²), and *large-format or campus sites* (> 20,000 m²). The median (6,120 m²) sits in the mid-size zone; the mean (18,849 m²) is pulled upward by the large campuses.

5.2 Breakdown by size bracket

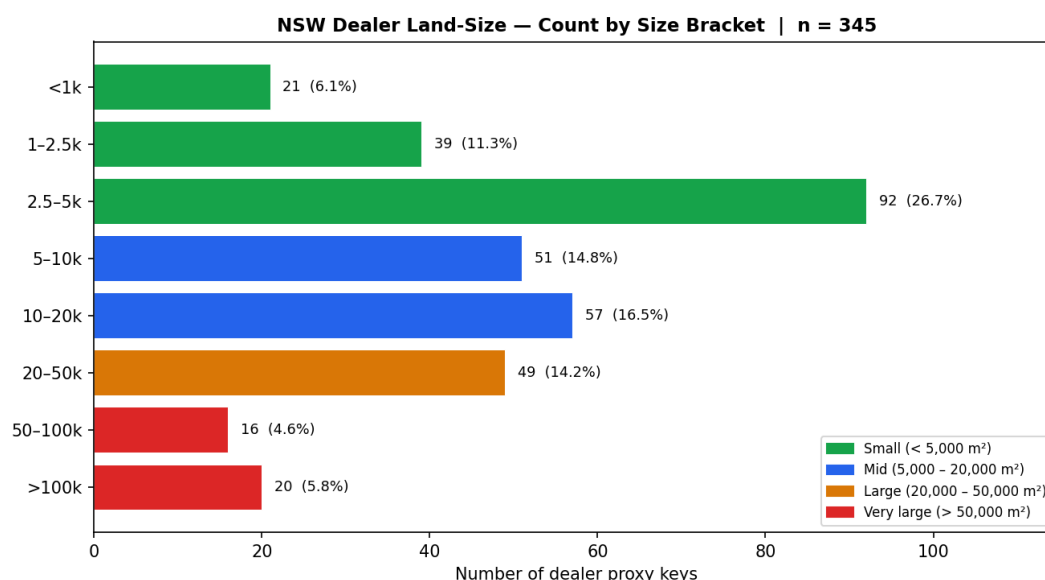


Figure 3: Dealer proxy key count by land-size bracket. The modal bracket is 2.5–5k m² (92 keys), consistent with a standard suburban dealership on an arterial road. The > 100k m² bracket (20 keys) are all multi-brand campuses or large regional super-sites.

Size bracket	Keys	%	Typical profile
< 1,000 m ²	21	6.1%	Prestige urban showrooms; inner-city locations
1,000 – 2,500 m ²	39	11.3%	Compact dealerships; high-density suburbs
2,500 – 5,000 m ²	92	26.7%	Modal bracket ; standard arterial-road dealer
5,000 – 10,000 m ²	51	14.8%	Mid-format; volume brands with service bays
10,000 – 20,000 m ²	57	16.5%	Large suburban dealer; service and used-car lot
20,000 – 50,000 m ²	49	14.2%	Large-format; often multi-brand or regional
50,000 – 100,000 m ²	16	4.6%	Super-site; typically a dealer group campus
> 100,000 m ²	20	5.8%	Multi-brand campus (Alexandria, Warwick Farm)
Total	345	100%	

5.3 Percentile distribution

Percentile	Land size (m ²)
10th	1,334
25th	3,297
50th (median)	6,120
75th	18,409
90th	55,482
95th	117,528
99th	209,570

The gap between the 75th percentile (18,409 m²) and the 95th (117,528 m²) highlights the step-change between a large suburban dealer and a genuine campus operation. The top 5% of sites by area are categorically different assets from the median dealer.

6 Largest Dealer Sites

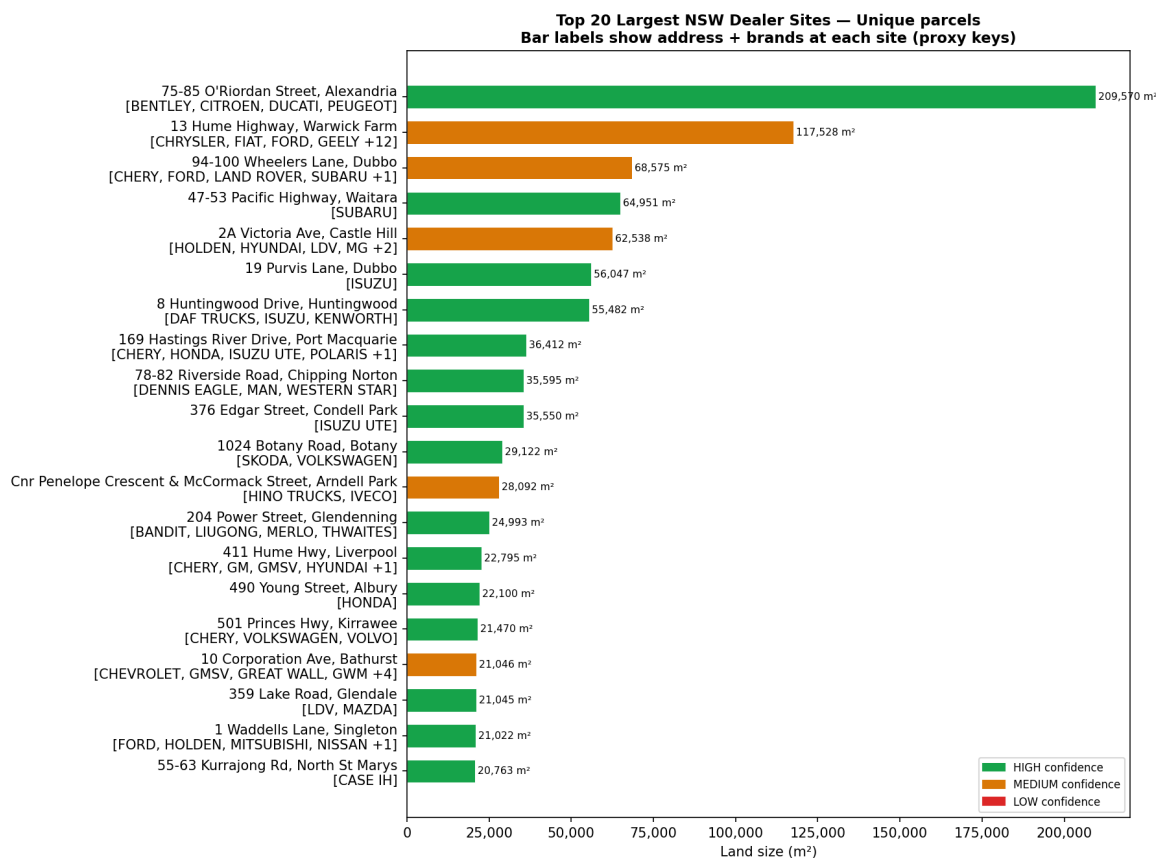


Figure 4: Top 20 largest unique dealer sites in the NSW pilot (address-deduplicated — each physical site appears once, regardless of how many brand proxy keys share it). Colour indicates confidence tier: green = HIGH, amber = MEDIUM.

Two sites dominate at a scale well above the rest of the distribution:

- **75–85 O’Riordan Street, Alexandria** (209,570 m²) — the Inchcape Australia / Trivett national campus. Houses Bentley, Citroen, Ducati, and Peugeot among others. This is the same complex classified as `OEM_HQ` for the Subaru, Kia, and Honda proxy keys; its retail-facing brands are included here.
- **13 Hume Highway, Warwick Farm** (117,528 m²) — the Peter Warren Automotive Group super-site. Eleven cadastral lots were summed via the NSW Property layer. Hosts Chrysler, Fiat, Ford, Geely, Honda, Isuzu UTE, and others.

Note on multi-brand sharing: each brand at the same site carries the same `LAND_SIZE_SQM` value. Do not sum across proxy keys to get a site-level total — deduplicate by `STREET_ADDRESS` and `SUBURB` first.

7 Land Size vs Vehicle Volume

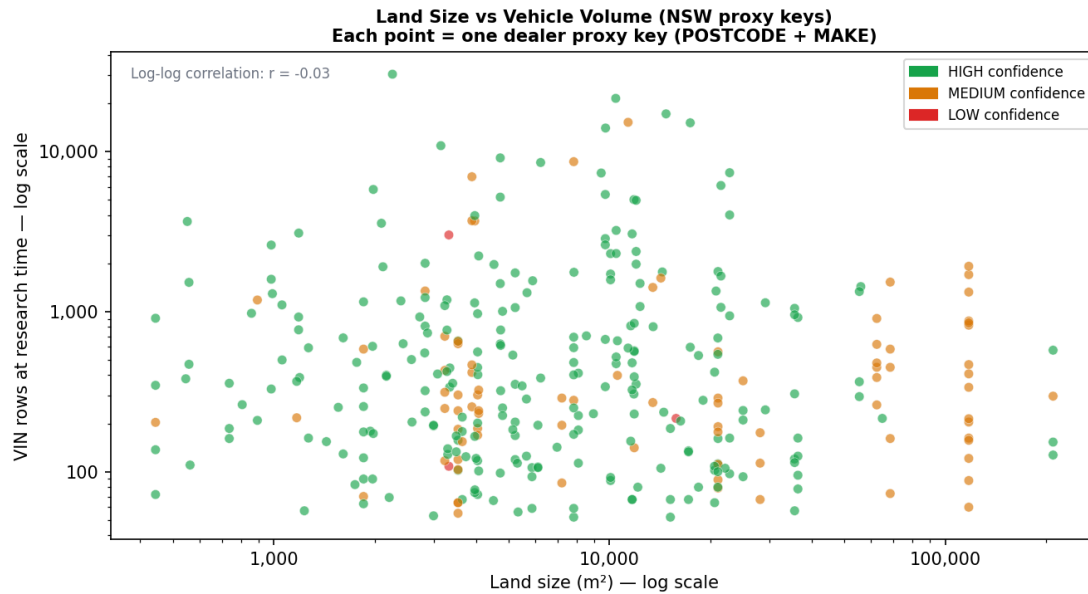


Figure 5: Scatter plot of land size vs. VIN rows (research-time snapshot) for each NSW proxy key. Both axes are log-scaled. Each point is one dealer proxy key (POSTCODE + MAKE). Colour indicates confidence tier. The log-log correlation is shown in the top-left corner.

The scatter shows a **weak positive log-log correlation** between site size and vehicle volume. This is expected: larger sites generally handle higher volumes, but the relationship is loose because:

- A high-value prestige brand may occupy a small showroom and sell fewer but more expensive vehicles.
- Urban inner-ring dealers often have constrained footprints but high throughput due to population density.
- Large regional sites may have extensive outdoor stock areas inflating parcel area without proportionately more transactions.

This is precisely why land size is valuable as a *contextual* input rather than a predictor of sales. The most analytically interesting dealers are those furthest from the trend line — either high throughput from a small site, or low throughput from a large one.

8 Make Coverage

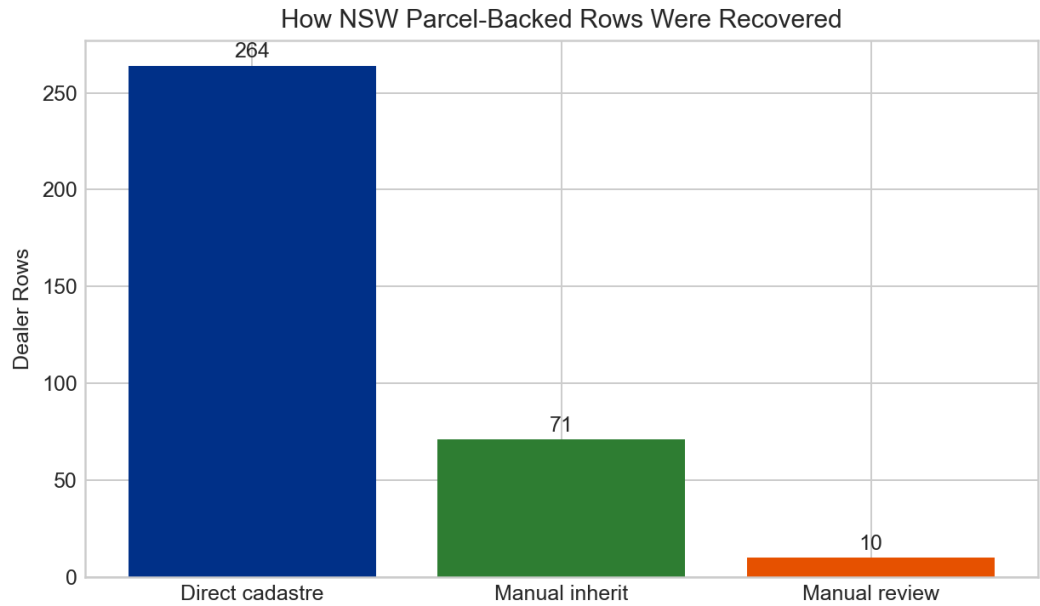


Figure 6: Recovery method breakdown by make for the top makes in the NSW pilot.

The 345 parcel-backed rows span **all major volume brands** in the NSW market. Ford (18 keys), Nissan (17), Mitsubishi and KIA (16 each), and Subaru (15) have the highest representation — reflecting both their dealer network density and the number of distinct postcodes in which they operate.

Prestige brands (Ferrari, Alfa Romeo, Bentley, Maserati) appear with smaller key counts but typically at the smaller end of the size distribution, consistent with compact urban showroom formats.

9 What Worked and What Remains

9.1 What worked

The key unlock was moving away from general-purpose web geocoding to the **official-source chain**:

Component	Why it mattered
G-NAF geocoder	Building-level precision for Australian addresses. Nominatim (OpenStreetMap) gave road-level matches for ~50% of dealer addresses; G-NAF resolved most of those.
NSW Cadastre WFS	Free, authoritative, CC-BY licensed. Direct point-to-parcel intersection with no interpretation layer — the <code>shape_Area</code> field is the official parcel area.
NSW Property layer	Essential for multi-lot campus sites. Warwick Farm's 11-lot Peter Warren campus would have been unresolvable without it.
Same-site inheritance	Multiplied the value of each confirmed site seed. One G-NAF + cadastre hit at Warwick Farm unlocked 10 inherited rows at the same site.

9.2 What is still missing

Gap	Keys	Path
<i>Geocode failures — NO_RESULT (102 rows total)</i>		
QLD (no geocode result)	67	Rerun <code>17_qld_gap_a_nominatim_recovery.py</code> off-peak, or use paid geocoder
NSW (no geocode result)	30	Hard address cases; manual review or Google Maps
VIC (no geocode result)	5	Manual address lookup
<i>Parcel failures — GEOCODED_REVIEW, no land size (114 rows)</i>		
All states (coordinates present)	114	Lat/lon available; parcel WFS returned nothing. Try wider buffer, road-type parcel logic, or same-site inheritance from nearby resolved rows
<i>Gap B (B013) — partial resolution: 207 / 292 resolved (done)</i>		
NSW (B013, resolved)	93 / 101	Pipeline completed
QLD (B013, resolved)	84 / 99	Pipeline completed
VIC (B013, resolved)	40 / 92	Pipeline completed
Remaining B013 unresolved	85	Included in geocode/parcel gap rows above
<i>Out of scope — no pipeline built yet</i>		
WA	138	Landgate WA (open under CC-BY)
SA	120	SA Spatial (open under CC-BY)
NT	25	NTLIS (access terms to confirm)
ACT	22	ACT Environment (open)
TAS	12	LIST Tasmania (open)

10 Recommended Next Steps

1. **NSW (91.6%) and VIC (88.2%) are ready for analysis.** Coverage at these levels is sufficient for dealer benchmarking and sales intensity analysis. QLD at 81.9% has improved significantly but still has 82 rows to resolve.
2. **Work through the 191-row master backlog (84_manual_review_master_backlog.csv)**
Rows are classified into three queues: 50 TARGETED_PARCEL_RERUN (automated, highest priority), 68 PARCEL_SELECTION_REVIEW (semi-manual), and 73 GEOCODE_COORDINATE_REVIEW (manual coordinate lookup). Tackle in that order for best effort-to-coverage ratio.
3. **Load LAND_SIZE_SQM into DIM_DEALER.** Run 18_dim_dealer_name_enrichment.py. This pushes all 1,530 east-coast rows into Snowflake DIM_DEALER. Add a UNITS_PER_1000SQM intensity metric to MART_DEALER_BENCHMARK_V for throughput benchmarking.
4. **Extend to remaining states (WA, SA, NT, ACT, TAS) when priority allows.** WA (138 keys) and SA (120 keys) are the highest-volume gap. All five states publish open cadastral data; only the endpoint and access token need wiring in.

11 Conclusion

The dealer land-size enrichment is a working analytical layer with **87.5% east-coast coverage (1,339 / 1,530 rows)**, built entirely on official government data sources and reproducible by re-running the pipeline scripts. NSW (91.6%) and VIC (88.2%) are effectively complete. QLD improved from 72.7% to **81.9%** across multiple recovery passes (B013, B014, manual same-site review). The 191 remaining rows are documented in a structured master backlog (84_manual_review_master_backlog.csv) with priority queues.

The key outputs are:

- 1,339 east-coast dealer proxy keys with parcel-backed land-size values in 34_dealer_land_size_east_coast_pilot.csv (1,530 rows total)
- The final dealer reference 71_dealer_reference_final.csv (2,142 rows) seeded into Snowflake DEALER_REFERENCE_T
- A reproducible pipeline (scripts 13–17) ready to extend to WA, SA, and remaining states

The enrichment connects the commercial performance data in MART_DEALER_BENCHMARK_V to the physical reality of dealer operations. Combined with sales volume and logistics metrics already built, the dataset now supports dealer efficiency analysis that was not possible from DMS data alone.

licence). Data is accurate as of April 2026.